DR. N. BHARATHIRAJA

Associate Professor | Ph.D. (CSE, Anna University) | Post-Doctoral Researcher | 15 Years Teaching Experience

rajamesoft@gmail.com | +91 8248361817 | Bangalore, India
0000-0003-4917-6691 | Google Scholar

TECHNICAL SKILLS

Python | C | Web Technology | Machine Learning | Deep Learning | Artificial Intelligence | Cloud Computing | IoT / WSN | Data Structures | Web Services | Problem Solving

RESEARCH METRICS

Scopus Papers: **56**
Scopus H-Index: **13**
WoS / SCIE: **22 papers**
WoS H-Index: **6**
GS H-Index: **15**
Scopus ID: 57224333742
WoS ID: C-2719-2018
ORCiD: 0000-0003-4917-6691

SOFT SKILLS

- Dedicated & committed educator
- Excellent programming & teaching
- Strong communicator
- Quick learner & good listener
- Self-motivated motivator

PHD RESEARCH

Topic:
Web Service Design for Efficient Data Sharing on Cloud with Wireless Sensor Network using Push Technology Architecture

University:
Anna University, Chennai
Awarded: August 2019

MEMBERSHIPS

IEEE Senior Member
ID: 98903965
ISTE Life Member
LM 141403
ACM Member
ID: 4798972
Internet Society
ID: 2228723
IAENG, Hong Kong
No: 125962

CAREER OBJECTIVE

Seeking a challenging position to utilize skills and abilities in the field of Teaching and Education, with a focus on professional growth while demonstrating resourcefulness, innovation, and flexibility.

EDUCATIONAL QUALIFICATIONS

Post-Doctoral Research Programme
Kuala Lumpur University of Science & Technology, Malaysia
Jan 2026 – Present

Ph.D. in Computer Science & Engineering
Faculty of ICE, Anna University, Chennai, India
Awarded August 2019

M.E. – Software Engineering | First Class (80.09%)
Anna University, BIT Campus, Tiruchirappalli
May 2010

B.E. – Computer Science & Engineering | First Class (74%)
Pandian Saraswathi Yadav Engineering College, Sivagangai
May 2008

HSC – State Board, TamilNadu | 78.3%
June 2004

SSLC – State Board, TamilNadu | 83.8%
June 2002

TEACHING EXPERIENCE (15 YEARS)

Associate Professor – CSE
Dayananda Sagar University, Bangalore
June 2025 – Present

- Approx. 6 months tenure; PG/PhD Supervisor, School of Engineering.

Associate Professor – CSE
Chitkara University, Punjab
March 2022 – June 2025

- Tenure: **3 years 2 months**.
- Mentor & Research Supervisor, UG students.
- Chitkara Excellence Awards 2023, 2024, 2025.
- PG Supervisor, BITS Pilani, Rajasthan Campus.

Assistant Professor – CSE
SRM Institute of Science & Technology, Ramapuram, Chennai
March 2021 – March 2022

- Tenure: **1 year**; CSI Student Branch Coordinator.

IFERP
PMIN90751236
STRA / Eurasia Research
STRA-M191011

- PhD Scholars registered: 02.

Assistant Professor – CSE

Saveetha Engineering College, Chennai

August 2010 – March 2021

- Tenure: **10 years 7 months**.
- Test Centre Administrator – TCS, Aptech, Satvat (7 years).
- Result Analysis Coordinator (Nov 2012 – Nov 2020).
- Student Mentor (Jun 2013 – Jun 2019).
- Exam Cell Coordinator (Aug 2010 – Aug 2017).
- Nodal Incharge, Anna University (May 2012 – May 2017).
- Deputy Warden, Boys Hostel (Jun 2011 – Jun 2014).

AWARDS

Chitkara Award 2025
Best Research Author
Rs. 77,800 – Apr 2025

Chitkara Award 2023
Top Performer
Rs. 59,612 – Dec 2023

Chitkara Award 2023
Journal Publications
Rs. 6,600 – Feb 2023

Best Research
Rs. 32,500
Saveetha EC (2019–20)

Best Teacher
100% result:
Info Retrieval & Multi-core
AY 2017–18

Best Teacher
100% result:
Operating Systems
AY 2015–16

Best Teacher
100% result:
Software QA & Prog.
AY 2014–15

Research Prize
Rs. 3,000 – AY 2012–13

ROLES & RESPONSIBILITIES

- **Research Supervisor & Mentor** for UG students, Chitkara University Punjab (March 2022 – present).
- **CSI Student Branch Coordinator**, Department of CSE, SRMIST Ramapuram Campus, Chennai.
- **Test Centre Administrator** for external online exams – TCS, Aptech Pvt. Ltd., Satvat Pvt. Ltd. (7 years, Saveetha Engineering College).
- **Result Analysis Coordinator**, CSE Dept., Saveetha Engineering College (Nov 2012 – Nov 2020).
- **Mentor** for UG students, Saveetha Engineering College (Jun 2013 – Jun 2019).
- **Exam Cell Coordinator**, Saveetha Engineering College (Aug 2010 – Aug 2017).
- **Nodal Incharge**, Anna University question papers & answer scripts (May 2012 – May 2017).
- **Deputy Warden**, Boys Hostel, Saveetha Engineering College (Jun 2011 – Jun 2014).

PATENT DETAILS (11 PATENTS – 1 GRANTED)

1. Smart Cart Management System for Enhanced Shopping Experience
   App: 202311043752 | Filed: 29-06-2023
2. System & Method for Managing Healthcare Data
   App: 202311047242 | Filed: 13-07-2023
3. An Energy Management System and Method Thereof
   App: 202311054437 | Filed: 14-08-2023
4. Dynamic Optimization & Real-Time Traffic Data System
   App: 202311054438 | Filed: 14-08-2023
5. Crop Yield Protection System and Method Thereof
   App: 202211060060 | Published: 22/12/2022
6. Face Recognition & Validation During Examination
   App: 202211070127 | Published: 09/12/2022
7. Blockchain Based Automated Auditing System
   App: 202211054679 | Published: 13/01/2023
8. Ambulance Booking Under Emergency Medical Services
   App: 202211074420 | Published: 30/12/2022
9. **[Patent Granted]** Detection of Brain Tumor
   App: 202211058363 | Published: 24/02/2023
10. Heterogeneous Data Storage with Deduplication
    App: 202211060074 | Published: 16/12/2022
11. Face Print Generation Using IoT
    App: 202041055664 | Published: 22/01/2021

ACTIVE REVIEWER

IEEE Access | IEEE Trans. Industrial Electronics | IEEE Trans. Circuits & Syst | IEEE Communications Standards | IEEE Design & Test | MDPI Sensors | MDPI Applied Sciences | MDPI Sustainability | MDPI Symmetry / Mathematic | Frontiers in Computer Science | Int'l J. Cloud Computing (Springer) | IET Microwaves & Antennas | IJERT (SCI-Indexed) | Int'l J. Web Engineering | Al-Bahir Journal for Engineering

Cricket – 1st
Staff Tournament
Saveetha EC (2011–12)

Quiz – 3rd Prize
Inter-College, PSYEC
Feb 2007

PHD SUPERVISION

DSU – School of Engg.
SRMIST Ramapuram
2 scholars registered
Chitkara University
BITS Pilani, Rajasthan

THESIS EXAMINER

Bharathiar University
Jun 2022
Narahari Narasimhaiah
Ph.D-CB-JUL2013-0124

Saveetha University
Jan 2025
Ravindhar N. V.
Reg: 161603076

BOOKS PUBLISHED

Insights into Algorithms and Data Structures through Dynamic Visualization
ISBN: 979-8-89496-009-8
SHINEEKS Publishers, USA
Apr 2024

Artificial Intelligence and Applications
ISBN: 978-81-976342-8-4
San Int'l Book Publication, India 2024

The Art and Science of Career Building in Big Data Visualization
ISBN: 979-8-89496-011-1
SHINEEKS Publishers, USA
Apr 2024

Designing Tomorrow's News Today With Web Tech Chronicles
ISBN: 979-8-88940-880-2
SHINEEKS Publishers, USA
Dec 2023

Crafting Expressive Art with Hand Gestures in Python
ISBN: 979-8-88940-878-9
SHINEEKS Publishers, USA
Dec 2023

Design and Implementation of a Web-Based Student Complaint System
ISBN: 978-1-63278-929-7
SHINEEKS Publishers, USA
Dec 2022

CERTIFICATIONS

AI Foundations for Everyone
Coursera, Nov 2024

Generative AI for Executives
Coursera, Jun 2024

From Excel to Power BI
Coursera, Jun 2024

AWS Cloud Essentials
Coursera, Jun 2024

Graphic Design
Coursera, Feb 2024

Effective Teaching – Social Media
NITTTR Chandigarh, Jun 2022

Game Dev with Scratch
Coursera, Oct 2021

Python for Data Science
Greyatom, May 2020

Blockchain & Cryptocurrency
Coursera, Jun 2020

Python & Java
SoloLearn, Jul 2019

RESOURCE PERSON

Expert Talk
"AI & ML for Industry 5.0"

SELECTED JOURNAL PUBLICATIONS

Total: Scopus-56 | WoS/SCIE-22 | GS H-Index: 15 | Scopus H-Index: 13

[1] Nagu, B. et al. "Enhancing environmental sustainability through real-time bio-waste detection using YOLOv6-CSP and relevance vector machine." *Measurement*, 2025.
[SCIE Q1, IF=5.6]

[2] Bharathiraja, N. et al. "Development of hybrid explainable AI with Swin Vision Transformer intrusion detection for securing VANETs." *Int. Journal*, 2025.
[SCIE Q2, IF=2.0]

[3] Balasubramanium, S., Nagu, B. et al. "Advanced predictive analytics for bio-waste management using YOLOv8-SPP in smart cities." *Scientific Reports*, 2025.
[SCIE Q1, IF=3.9]

[4] Nagu, B. et al. "Enhancing Road Safety via Wearable Sensors and Explainable AI: Cognitive Load Monitoring." *Int. J. Intelligent Transportation Systems Research*, 2025.
[WoS Q2, IF=2.1]

[5] Praveen M., Dhandapani S., Bharathiraja N. "Optimizing Motion in Nanoscale Robotics: Hybrid WQPSO-Fuzzy Logic Approach." *Studies in Informatics and Control*, 2025.
[SCIE Q3, IF=1.1]

[6] Talari, P., Nagu B. et al. "Hybrid feature selection for early prediction of Type-2 Diabetes." *PLOS ONE*, 2024.
[SCIE Q1, IF=3.7]

[7] Sulaiman, A., Nagu, B. et al. "AI-Based Secured Power Grid Protocol for Smart City." *Sensors (MDPI)*, 2023.
[SCIE Q1, IF=3.9]

[8] Marappan, R., Kaur, G., Bharathiraja N. et al. "Efficient evolutionary modeling for lifetime maximization of WSN healthcare networks." *Soft Computing, Springer*, 2023.
[SCIE Q1, IF=3.7]

[9] Umamageswari, A., Bharathiraja N. et al. "Novel Fuzzy C-Means Chameleon Swarm Algorithm for plant disease classification." *ICT Express*, 2021.
[SCIE Q1, IF=4.754]

[10] Jayanthi, E., Bharathiraja N. et al. "Cybersecurity to detect credit card frauds in healthcare using new ML strategies." *Soft Computing*, 2023.
[SCIE Q1, IF=3.7]

[11] Bharathiraja, N. et al. "Elite Oppositional Farmland Fertility Optimization for node localization in wireless networks." *Wireless Comm. and Mobile Computing*, 2022.
[SCIE Q2, IF=2.146]

[12] Kaur, G., Bharathiraja N. et al. "Secured intelligent transportation with privacy retention through blockchain." *J. Intelligent & Fuzzy Systems*, 2024.
[SCIE Q3, IF=2.0]

SELECTED FDPS & WORKSHOPS (41 TOTAL)

- 5-Day FDP on AI in Engineering – Chitkara University (Jun 2023)
- 1-Week FDP "Multiverse Approach" – SRM Madurai CET (Aug 2023)
- 3-Day FDP on ML & Computer Vision – VIT (Nov 2021)
- AICTE STTP on Blockchain & ML – KLN College of Engineering (Apr 2021)
- AICTE-ATAL FDP on Blockchain – KCG College, Chennai (Sep 2020)
- AICTE-ATAL FDP on IoT – Dr. Mahalingam College, Coimbatore (Sep 2020)
- FDP on Social Network Analysis – Anna University Chennai (Nov 2019)
- ISTE Workshop on Computer Programming – IIT Bombay (May–Jun 2014)
- FDP on Cryptography & Network Security – Panimalar IT (May 2016)

SRM Madurai CET
Jul 2024

National Seminar
Applied Soft Computing
SRM Madurai CET
May 2023

Lecture
"Python Problem Solving"
SREC, Coimbatore
Nov 2022

- Cloud-based ML for Big Data Mining – Anna University (Feb 2015)
- Mentor, Smart City Rajkot HACKATHON 2017, Gujarat